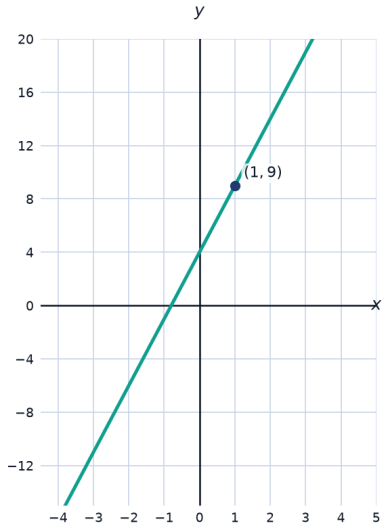
Radicals & Rational Exponents

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

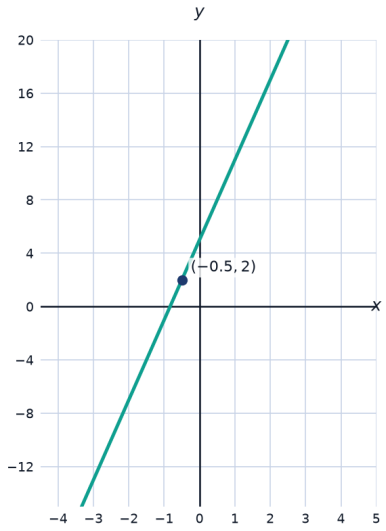
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(2)$.



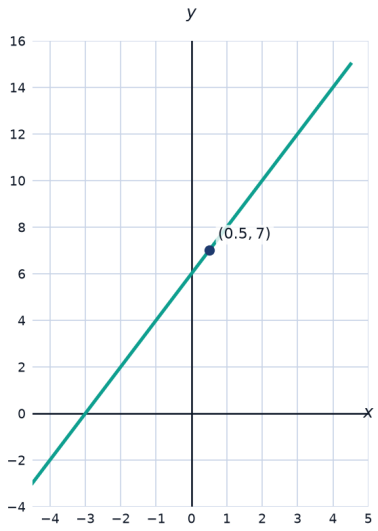
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for logarithmic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(2)$.

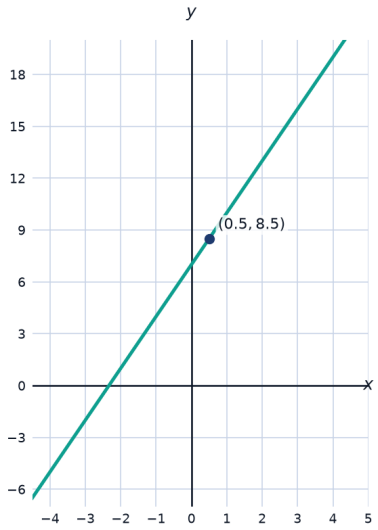


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for extension: properties of logarithms consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

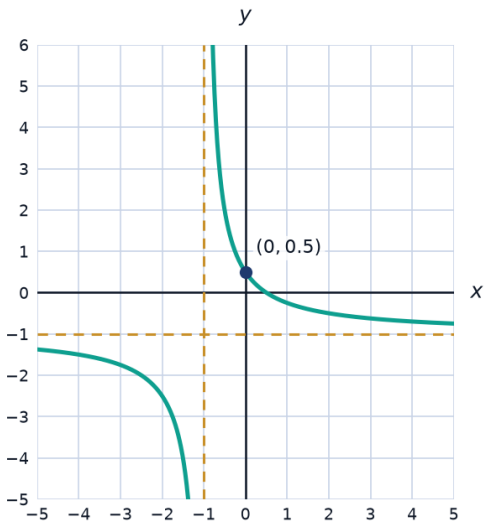
Solve the equation $\sqrt{2x+7} = x+2$ for all real values of x.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Radicals & Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in x?

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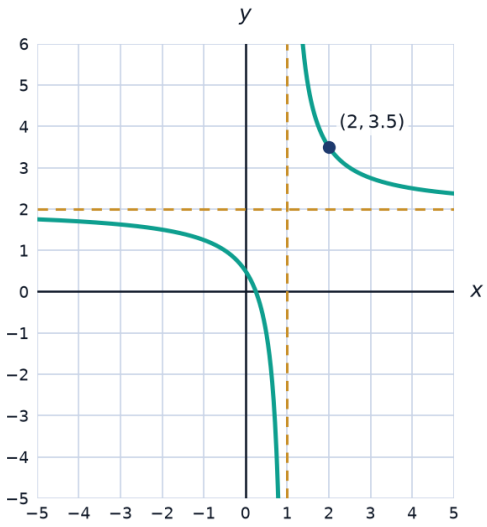
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Matrices & Systems

A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability & Statistics

[AP Precalculus Exam Format & Modeling Framework · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

[Unit 1 · Polynomial and Rational Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

[Exponents & Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

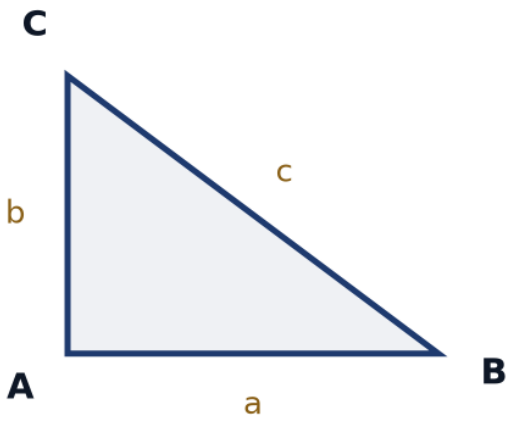
19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

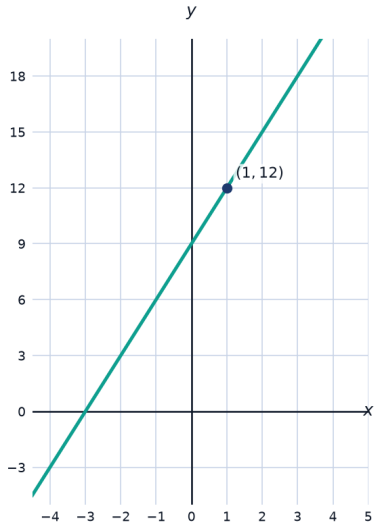
In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.



a. 15

b. 18

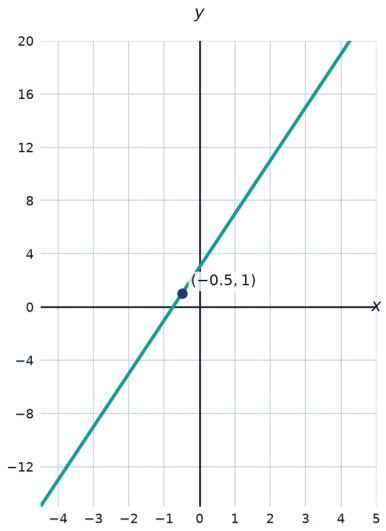
c. 6

d. 19

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(2)$.



a. 11

b. 15

c. 8

d. 15

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for applications of exponential functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For a constant k, the equation $\sqrt{2x+k} = x-1$ has $x=5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?

a. -5

b. -1

c. 1

d. 6

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the equation $\sqrt[3]{25-x^2} + \sqrt[3]{x^2+3} = 4$ What is the product of all real solutions to this equation?

a. -24

b. 24

c. -48

d. 0

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $g(x) = \sqrt{x^2-6x+9} + \sqrt{x^2+8x+16}$. Find the minimum possible value of $g(x)$.

a. 2

b. 3

c. 4

d. 7

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the value of the expression $\sqrt{36} + \sqrt{49}$?

a. 10

b. 11

c. 13

d. 15

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Check: is $x=-1$ a solution of $\sqrt{x} = -1$?

a. Maybe

b. Only in $\mathbb{C}$

c. No (square root is non-negative)

d. Yes

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write $x^{3/2}$ as a radical.

a. $x^{1/2} \cdot x$

b. $x\sqrt{x^2}$

c. $\sqrt{x^3}$ or $(\sqrt{x})^3$

d. $\sqrt[3]{x^2}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find $f(x) = \sqrt{x} + \sqrt{1 - x}$. What is the domain?

a. $x \geq 0$

b. All reals

c. $0 < x < 1$

d. $0 \leq x \leq 1$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Write $\sqrt[3]{a^2 b}$ using rational exponents.

a. $a^{2/3} b^{1/3}$

b. $a^{1/2} b^{1/3}$

c. $a^{3/2} b^{1/3}$

d. $a^2 b^3$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following equals $8^{-2/3}$?

a. $\frac{1}{8}$

b. 4

c. $\frac{1}{4}$

d. -4

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the real number(s) equal to $(-32)^{3/5}$.

a. 8

b. Undefined

c. -8

d. -32^3

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Write $\sqrt[4]{x^3}$ using rational exponent.

a. x^{3+4}

b. $4x^3$

c. $x^{4/3}$

d. $x^{3/4}$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\sqrt{4}$ — note: solve $\sqrt{36}$.

a. 9

b. 6

c. 3

d. 12

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the number of real solutions of $\sqrt{x} = x^2 - 6x + 9$.

a. 0

b. 1

c. 4

d. 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

Solve: $\sqrt{2x + 3} = x$. Check for extraneous roots.

a. $x = 3$ only

b. $x = -1$ only

c. $x = 3$ and $x = -1$

d. No solution

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

Prove $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ using rational exponents.

a. True only for $n = 2$

b. $(ab)^{1/n} = a^{1/n}b^{1/n}$ — follows from power rules for real numbers

c. Not always true

d. Requires $a, b > 0$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Simplify: $\frac{\sqrt{x+h}-\sqrt{x}}{h}$ (rationalize numerator).

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x+h}+\sqrt{x}}$

c. $\frac{h}{\sqrt{x+h}+\sqrt{x}}$

d. $\frac{\sqrt{x+h}-\sqrt{x}}{h}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Express $\sqrt[6]{x^4}$ in simplest radical form.

a. $\sqrt[3]{x^2}$

b. $\sqrt[6]{x^4}$

c. $x^{2/3}$

d. Both A and C

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Conic Sections

Simplify $\frac{5+\sqrt{3}}{3-\sqrt{3}}$ by rationalizing.

a. $\frac{8+6\sqrt{3}}{6}$

b. $\frac{15+5\sqrt{3}+3\sqrt{3}+3}{6}=\frac{18+8\sqrt{3}}{6}=3+\frac{4\sqrt{3}}{3}$

c. $5\sqrt{3}$

d. $\frac{5+\sqrt{3}}{3}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Trigonometric Functions

Find the real solutions of $x^{2/3}=2$.

a. $x=n^{n+1/n}$

b. $x=n+1$

c. $x=n^2$

d. $x=n^{n/n+1}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

Prove by induction that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Step: assume true for n, verify for $n+1$.

a. Not provable by induction

b. Only for $n \le 100$

c. $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓

d. Only for even n

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

Determine the range of $f(x) = \frac{x}{x+1}$.

a. $0 \le y < 1$

b. $-1 < y < 1$

c. All reals except $y = 1$

d. All reals

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